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SEMI E187 Compliance Reference Guide

The Asset Security Life Cycle

TECHNICAL MARKETING TEAM
TXOne Networks Inc.

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Executive Summary

Hard-hitting cyber attacks on organizations have become commonplace in recent years, and the advanced manufacturing industry has not been an exception. Semiconductor manufacturing in particular has been under pressure to develop better production security due to the critical importance of their output and high degree of automation. In January of 2022, SEMI took the next step and launched their new SEMI E187 cybersecurity standards, which were designed to neutralize supply chain attacks, insider threat, and other potential flash points in a cyber attack.

However, many semiconductor specialists are struggling to find a practical approach to putting SEMI E187 into practice, from establishing new cybersecurity routines to implementing better technical control methods. This paper proposes basing defensive oversight and planning on the asset life cycle, thereby securing factory equipment by defining and monitoring both expected behaviors and necessary routines during onboarding, staging, production, and maintenance. This paper will also summarize the key challenges of the SEMI E187 standards and provide some practical recommendations to support asset owners and equipment suppliers in their journey towards compliance.

Introduction

- **Critical Properties of Cybersecurity for OT Systems**

Many OT (operational technology) environments are now equipped with operating systems and networking technologies that originated in IT (information technology). In turn, this has rendered OT systems vulnerable to break-in techniques that hackers originally developed to take down IT systems.

Malware is often found in devices such as USBs and laptops carried by personnel (so-called 'insider threat'), and has even been found hidden in new assets arriving for onboarding ('enemy asset' supply chain attacks). Infected devices or equipment brought in by personnel, maintenance technicians, or outside contractors that can interact with OT computers and networks have become a typical attack vector for attackers attempting to instigate a cyber incident.

As in any OT environment, within the semiconductor foundry safety is the most important concern. Safety is then first followed by availability, second by integrity, and third by confidentiality (sometimes called the "AIC" model). Conversely, confidentiality is the priority in IT, followed by integrity, with availability last (the "CIA" model). Generally, semiconductor manufacturers will struggle with long equipment life cycles, deploying patches, and meeting strict requirements for availability and performance, which they must balance against cybersecurity concerns. Security practices must therefore be designed to unceasingly uphold the AIC priority model.

- **Long Equipment Life Cycle**

In semiconductor manufacturing, the life cycle of production equipment is generally more than 20 years. In this case, there are bound to be many types of equipment with legacy operating systems inside the factory. Furthermore, on average every work site can expect at least a few assets or the software they operate to pass their EOL (End-of-Life) date each year. These legacy OSes will no longer receive vendor updates or security patches, and without defenses informed by this knowledge they will be exposed to steadily increasing cyber risk as new vulnerabilities are discovered that will never receive an official patch.

- **Difficult to Patch**

For those assets that are still receiving regular updates from manufacturers, applying patches can be an arduous task. To patch any asset in an operational environment, asset owners must balance compatibility issues with dependencies ("Is the patched asset compatible with the network and other assets?"), find a maintenance window ("When is the right time to conduct this update so that it will not disrupt output?"), and reach consensus with other stakeholders. Asset owners in a semiconductor fab must carefully take compliance issues into account to ensure the continuity of production processes, high availability, and the avoidance of disruptions to production.

- **Availability, Performance, and Quality Must Come First**

Traditional IT management methods present a significant challenge to cybersecurity personnel on an OT work site. As the original configurations and setup on equipment are changed to match the environment, equipment performance and the asset's warranty can be jeopardized. To prevent cyber incidents and meet the complex challenges of the OT environment, TXOne's specialists have collaborated with industry leaders to create a new approach to management based on applying OT zero trust to the asset life cycle. By zeroing out trust and implementing a policy of verification at every phase that could present a potential point of entry to a threat, cybersecurity specialists can streamline compliance with SEMI E187, address OT cybersecurity challenges, and secure fab equipment against cyber threats.

Overview of the Asset Security Life Cycle

- **Achieving OT Zero Trust Throughout the Asset Security Life Cycle**

An owner of production equipment in a semiconductor foundry usually begins by evaluating existing critical assets and the supporting services required for operation. To analyze the demand for assets, they will also determine management strategies. In this case, OT zero trust-based strategies have been found effective during the asset planning phase, where ensuring production resilience is critical.

The premise behind achieving OT zero trust is that it is not safe to trust anything inside or outside the OT environment, including stakeholders, the network, and assets, without first identifying and classifying all users and devices seeking access. The goal of OT zero trust is to eliminate all threats, whether they originate from outside or within the network. Adopting this strategy is critical to securing the operational resilience of OT systems while preventing potentially costly cyber incidents. This section will show how the asset life cycle architecture can be used as a framework to implement down-to-earth OT zero trust-based defenses for modernized factories.

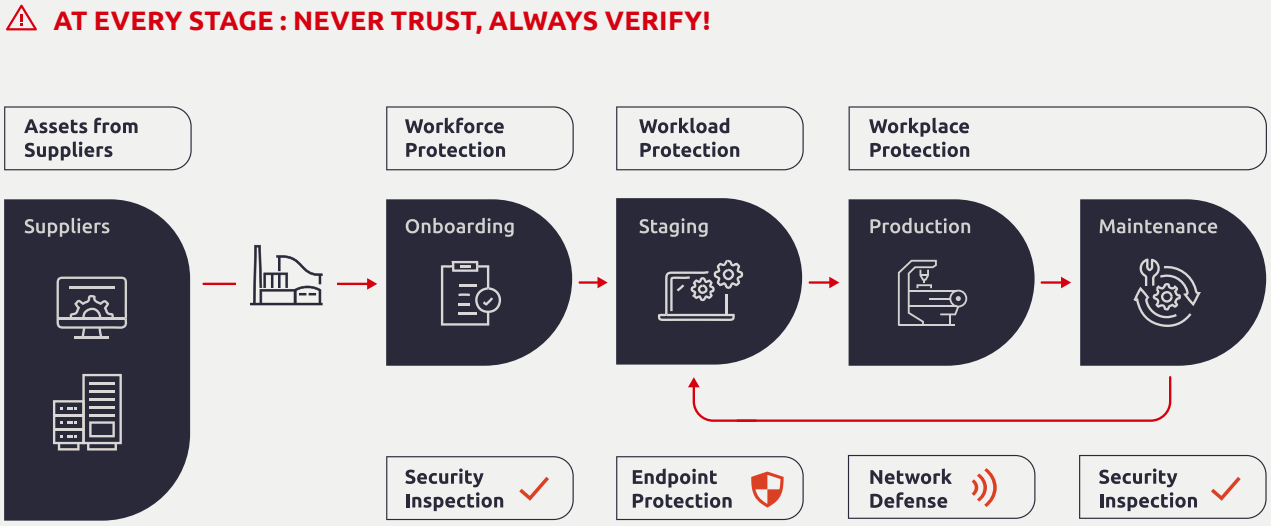


Figure 1, guiding principles of OT zero trust organized by the asset security life cycle

▪ Guiding Principles of the Asset Security Life Cycle

Understanding the protection of the asset life cycle is critical for realizing OT zero trust strategy. This section will classify and describe four critical stages to protect the asset life cycle: onboarding, staging, production, and maintenance.

1. Onboarding

Before an asset is shipped to your facility, suppliers should scan each asset and create a record of OT health that proves the device is malware-free. Similar to going through each country's customs on either side of an international flight, both the supplier and the asset owner must independently confirm device safety and security for themselves on their own side of the transaction. As each piece of equipment arrives at your facility, it must therefore be treated as hostile until it's been scanned for threats and any potentially exploitable vulnerabilities have been documented.

Any asset, even one that's brand-new from the manufacturer, has the potential to deliver a threat or introduce a convenient back door for hackers to your operational environment. Given time, vulnerabilities will be found in each asset over its lifespan that must be addressed to maintain operational integrity. Mitigate security risk and prevent 'enemy assets' from connecting to the internal network by inspecting all equipment as it arrives. Use security inspection to guarantee that each device entering your work site is fit for use, as well as to identify vulnerabilities that will need to be secured for the asset's life span.

2. Staging

Staging is the process of locking down assets to eliminate means of attack. This includes patching vulnerabilities and shutting down non-essential services such as system applications, permissions, ports, user accounts, and other features that are not needed. By locking down assets, technicians minimize opportunities for attackers to gain access to mission-critical computers and stop malicious applications from running. We can achieve this goal by following these practices:

- To protect each system, deploy an anti-malware solution such as trust lists which only allow listed applications and operations to execute
- Remove superfluous services and software
- Disable high-risk network protocols and unnecessary network ports
- Control and limit user privileges and access
- Disable USB access pending administrator approval

3. Production

Anything that goes wrong during the production stage will cause immediate financial damage, something asset owners are acutely aware of as they design policies and practices to accommodate it. Protecting as-

sets during the production phase begins with steps taken to configure and set up the asset during staging. As the asset enters the production phase, network security is the new variable that must be meticulously secured to guarantee operational integrity. This means taking into account the needs of both modernized and legacy assets so that productivity-based network segmentation respects and safeguards their work.

Network segmentation is an advanced OT zero trust-based methodology whereby different assets on the operational network are more efficiently and conveniently segmented using next-generation firewalls or IPSes to strictly limit user and asset privileges. Assets are grouped into each segment based on their work needs, and then specialized policies are applied to each segment to caretake the special needs of the assets within. Asset owners must be prepared to defeat a wide range of cyber threats that attackers are eager to spread through the network, and segmentation makes this possible with improved network access controls and better analytics. Network segmentation prevents one compromised asset from turning into a large-scale disaster, streamlines oversight, and makes it much more challenging for hackers to gather information or move within your network.

4. Maintenance

From the moment an asset is put into its intended production use, it begins to age and depreciate and the regular cycle of maintenance begins. Maintenance means not only repairs but also software configuration changes, system upgrades, and security updates. Through these routines, technicians keep assets in sync with the most advanced cybersecurity protections, ensure compliance with company security policies, and extend asset life and value.

Routine malware scans are a fact of life for OT assets, and every time there is a change to software or configurations it's time for another scan. Locked down assets are routinely checked for malware by their lockdown software, but what about assets that are standalone or highly sensitive? The best way to cover these special cases is a handheld rapid scan device that requires no installed software, so as to leave the scanned device's software and configurations totally undisturbed but verified clean, and centralizes logs to a single pane of glass for archiving and reference. With the rapid scan complete and the results logged, the secured and maintained asset is ready to go through staging and back into production.

Mapping the Asset Security Life Cycle to SEMI E187

Faced with the security challenges that come with the operation and maintenance of complex smart factory equipment, CISOs must strike a balance between cybersecurity and economics.

For example :

- How can asset life be extended rather than retiring the asset?
- How can downtime be avoided or minimized during security patch installation?
- How can cybersecurity be maintained without interrupting operations?

Production equipment is delicate, especially in the semiconductor industry.

Without the support of equipment suppliers, it is not easy to achieve a complete asset life cycle protection to realize OT zero trust strategy. Thus, in January 2022, SEMI released a new security standard: SEMI E187 defines cybersecurity baseline requirements to reduce malware from spreading to the factory during initial equipment onboarding and throughout ongoing production activities, including field service repairs, software patching, and maintenance (refer to table 1). Obviously, equipment suppliers must clearly understand customer policies. The supplier and the customer must cooperate with each other to build safe and resilient smart factories.

Table 1: Overview of SEMI E187

<table>
<tr><td rowspan="9">System Level</td><td>Inter-Enterprise</td><td></td><td></td><td></td><td></td></tr>
<tr><td>Enterprise</td><td></td><td></td><td></td><td></td></tr>
<tr><td>Factory</td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="2">Equipment (Production equipment and AMHS equipment)</td><td colspan="2">Equipment suppliers enable equipment to support the requirements of SEMI E187</td><td colspan="2">Asset owners use SEMI E187 as a baseline for equipment cybersecurity management. Equipment suppliers can provide appropriate support if required.</td></tr>
<tr><td colspan="4">Focusing on computing devices installed with Windows or Linux operating systems</td></tr>
<tr><td>Module</td><td></td><td></td><td></td><td></td></tr>
<tr><td>Subsystem</td><td></td><td></td><td></td><td></td></tr>
<tr><td>IO Device</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="2" rowspan="2"></td><td>Onboarding</td><td>Staging</td><td>Production</td><td>Maintenance</td></tr>
<tr><td colspan="4">Lifecycle Phases</td></tr>
</table>

Key Concept of SEMI E187

We found that this standard explicitly focuses on improving the cybersecurity of the operating system, network, and endpoint to facilitate cybersecurity management of semiconductor manufacturers.

- **Computer Operating System Security**

Support for End-of-Life Operating Systems

End-of-Life (EOL) refers to the point in time when the developer will cease to create updates and patches for a piece of software. Running an operating system that has reached EOL, a so-called "legacy" operating system, exposes the work environment to significantly more cyber risk than a vendor-supported operating system. Common cyber risks include:

- New vulnerabilities: When an operating system vendor stops issuing updates and patches, the operating system becomes orders of magnitude more vulnerable to cybersecurity threats.

- Asset incompatibility: Equipment suppliers and users may be concerned about the security patch of the operating system affecting overall equipment performance due to incompatibilities between the updated operating system and applications or other equipment side effects.

Using vendor-supported operating systems decreases the likelihood of introducing additional security vulnerabilities. If the equipment supplier finds that the software has reached its EOL, ideally they will not use it in the design and development process because of its inherent low security.

When the equipment's operating system has a critical security update and is under a maintenance contract, an ideal equipment supplier should proactively provide documentation to guide asset owners on how to apply security patches to equipment. In the best case, the equipment supplier should also conduct software compatibility, software package dependency, performance impact, and side effect evaluation for security patches and provide the analysis results to asset owners for decision-making. Asset owners can use the results of this analysis to decide whether to update, thereby balancing cyber risk and operational performance.

In other words, one day the operating system will still be EOL, but it is often necessary for asset owners to continue using an OS that has passed EOL for extended periods to support the business mission. This timeline may extend well beyond the support offered by the equipment supplier, and as vulnerabilities remain unpatched, they accumulate cyber risk. To mitigate these risks, asset owners should assess EOL status and deploy virtual patching as necessary. For an asset that still receives updates virtual patching can act as a temporary care measure until the patch is available, while for legacy assets outside of their EOL date it can provide long-term shielding. Deploy OT-native virtual patching to prevent attackers from exploiting vulnerabilities to compromise your operation.

Recommendations :

- Track operating system EOL status
- Deploy virtual patching to mitigate the risks of unsupported or unpatchable operating systems

- **Network Security**

Network Transmission Security

If the production equipment uses web service applications, file transfer services, or terminal services (telnet), equipment suppliers should support the use of secure protocols such as HTTPS, SFTP and SSH to allow asset owners to transfer files and manage their network credentials. For example, HTTP has been one of the foundations for data communication in web services, but unfortunately, it broadcasts its requests in the clear instead of securing the data through an encrypted channel. While it may be difficult to completely rule out the use of HTTP, we suggest that equipment should support secure transmission protocols such as HTTPS to provide asset owners with a security option to implement this transition on demand.

TXOne's own Edge series of next-generation firewalls and IPSes provides network visibility so that asset owners can know which protocols are in use. The Edge series can understand and make specific rules for the majority of OT protocols so that when a normally insecure protocol is critical to a piece of equipment, trust lists can be created based on specific protocols or specific commands within a protocol.

Network Configuration Management

The asset owner needs to restrict the use of all unnecessary ports, protocols, and system services to limit entry points available to attackers. For example, SMB Session (Server Message Block), which runs on port 445, has some capabilities such as computer file and printer sharing that have long been known to be a favorite target of attackers when exposed to the network.

However, sometimes risky network ports such as port 445 are required in internal networks and systems to do specific tasks. In this case, the equipment vendor will need to suggest compensative cybersecurity controls at the network layer, which is required to mitigate cyber risk. Ideally, the equipment suppliers should provide a document to the asset owner on managing equipment network configuration methods, such as enabling or disabling network protocols or ports.

Recommendations:

- Support secure transmission network protocols and services
- Provide a method to disable unnecessary network ports and protocols
- Provide a method to changing the network configuration and port usage should be described if applicable

- **Endpoint Protection**

Vulnerability Mitigation

As an old security mantra goes, "You can't protect what you can't see." Equipment vendors should use every means available including vulnerability scanning tools to find and resolve critical vulnerabilities before shipment to reduce the chance of vulnerabilities being exploited.

Malware Scanning

It is equally essential to scan equipment for malware before shipment. Equipment vendors should provide a malware-free certification that their equipment has passed antivirus scanning with no viruses, spyware, or malware found. It is worth noting that malware changes daily or sometimes hourly. The scanning tool must record the date and details from the scan in the certification to keep scans effective.

Anti-Malware Protection

Anti-malware solutions include, but are not limited to, antivirus applications that deploy allow lists for control mechanisms. SEMI E187 expects equipment suppliers to provide end users with the ability to install, manage, or update anti-malware mechanisms that can mitigate the effects of malicious code or unauthorized software. The system must also provide the capability to update protection mechanisms as necessary.

Security Hardening

Some of the functions and services provided by default may not be necessary to support essential operations in the fab equipment. Equipment suppliers should consider disabling unused or unnecessary physical ports, logical ports, and protocols to prevent unauthorized connection to devices, unauthorized transfer of information, or unauthorized tunneling.

Access Control

Under SEMI E187 equipment vendors should provide authentication methods such as passwords, tokens, and biometrics to ensure identity authentication. Multi-factor authentication should be used when possible because it is more secure. In addition to having identity authentication, equipment vendors should also consider the segregation of duties of each user account. At least separate administrator accounts from standard accounts. Provide a management method that meets the principle of least privilege for user accounts with specific responsibilities in order to prevent unauthorized access.

Recommendations:

- Conduct vulnerability scans before shipment, creating a scan log that includes date and time
- Conduct malware scans before shipment to create a dated scan report that proves equipment is malware-free
- Evaluate the compatibility of at least one software application that can protect systems from malicious files and scripts

- Choose a method of security hardening
- Implement user authentication methods for operating systems and applications
- Implement segregation of duties and the principle of least privilege for operating systems and applications

- **Security Monitoring**

Security event logs provide essential information about system operation. This information aids in error correction, continuous monitoring, and incident investigation. The equipment should be able to generate and transmit security event logs, which should record the event type, event description, user account, and timestamp to prevent security breaches, mitigate the impact of security events, and comply with system security policies. Furthermore, security event logs should be protected and access should be limited to authorized individuals. Above all is to have security event logs available to personnel as necessary to maintain and operate assets and cyber defenses.

Recommendations :

- Record and archive system and application security event logs
- System event logs should include records of access control, configuration changes, and system errors at minimum
- Keep security event logs secured
- Aggregate security event logs
- Provide tools to monitor and review security logs
- Normalize security event logs in a machine-readable format

Table 2: SEMI E187 Concept Mapping with Asset Lifecycle Protection

<table>
<tr><th>Stage in the Asset Security Life Cycle</th><th>SEMI E187 Requirement</th><th>Technique to Satisfy Requirement</th><th>The OT Zero Trust Solution</th></tr>
<tr><td rowspan="3">Onboarding</td><td rowspan="2">Computer Operating System Security Requirement</td><td>End-of-Life OS check</td><td rowspan="3">• Trend Micro Portable Security 3™</td></tr>
<tr><td rowspan="2">Endpoint Protection Requirement</td><td>Vulnerability scan</td></tr>
<tr><td>Malware scan</td></tr>
</table>

<table>
<tr><th>Stage in the Asset Security Life Cycle</th><th>SEMI E187 Requirement</th><th>Technique to Satisfy Requirement</th><th>The OT Zero Trust Solution</th></tr>
<tr><td rowspan="3">Staging</td><td rowspan="2">Endpoint Protection Requirement</td><td>Antivirus software / application trust listing enabled on equipment</td><td rowspan="2">▪ StellarProtect™
▪ StellarEnforce™</td></tr>
<tr><td>Equipment can disable unnecessary capabilities</td></tr>
<tr><td>Computer Operation System Security Requirement</td><td>Patch management (via virtual patching)</td><td rowspan="5">▪ EdgeIPS™
▪ EdgeFire™
▪ OT Defense Console™</td></tr>
<tr><td>Network Security Requirement</td><td>Network configuration management</td></tr>
<tr><td rowspan="5">Production</td><td>Network Security Requirement</td><td>Equipment supports encrypted protocols</td></tr>
<tr><td rowspan="2">Security Monitor Requirement</td><td>Recording and exporting security event logs</td></tr>
<tr><td>Complete visibility of asset, operational, and security events</td></tr>
<tr><td rowspan="2">Endpoint Protection Requirement</td><td>Antivirus software / application trust listing enabled on equipment</td><td rowspan="2">▪ StellarProtect™
▪ StellarEnforce™</td></tr>
<tr><td>Equipment can disable unnecessary capabilities / Application-level access control</td></tr>
</table>

Stage in the Asset Security Life Cycle	SEMI E187 Requirement	Technique to Satisfy Requirement	The OT Zero Trust Solution
Maintenance	Endpoint Protection Requirement	Vulnerability scan	▪ Trend Micro Portable Security 3™
		Malware scan	
		System hardening checks	

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- **Security Inspection**: **Trend Micro Portable Security 3**

Trend Micro Portable Security 3™ makes it much easier for ICS owners and operators to scan for malware and to collect asset information on stand-alone computers as well as air-gapped systems. Unlike traditional anti-virus software, Portable Security scans and cleans up malware without installing scanning software, showing status with an easy-to-understand LED display. During the scan, Portable Security also collects asset information, helping to improve OT visibility and to eliminate shadow OT.

The Portable Security management program can deploy scanning configuration settings to multiple scanning tools, either remotely or physically. It also compiles and integrates the scan logs and asset information from multiple scanning tools in multiple locations, providing a holistic view of all endpoints. Trend Micro Portable Security 3™ Pro additionally comes equipped for secure file transfer, allowing end users to mitigate insider threat to air-gapped and standalone environments by transporting sensitive data with its 64GB of AES-256 hardware encrypted storage.

Table 3: Example of Inspection Solutions for SEMI E187

Key Solution		Description	SEMI E187 Requirement
1	EOL OS Scan	Allows users to perform OS information checking and documentation	OS version checking and documentation

Key Solution		Description	SEMI E187 Requirement
2	Vulnerability Scan	Allows users to perform on-demand vulnerability scanning without installing scan software on equipment	Vulnerability scanning and documentation
3	Malware Scan	Allows users to perform on-demand malware scans without installing antivirus software on the equipment	Malware scanning and documentation
4	System Hardening Checks	Allows users to check and document information on system hardening	Advanced feature
5	Centralized Management of OT Asset Visibility	View the status and event logs of all endpoints on a single pane of glass, streamlining compliance with requirements put forth by the SEMI E187 standards	Advanced feature

- **Endpoint Protection: The Stellar Series**

TXOne StellarProtect™ is the world's first all-terrain protection for modernized endpoints, a one-of-a-kind defensive solution custom-engineered for operational technology. Its advanced threat scanning brings known attacks to heel while its next-gen machine learning engine blocks unknown threats, requiring no internet access. ICS filtering on StellarProtect™ is based on an inventory of applications and certificates. This inventory allows it to verify trusted programs, eliminating unnecessary overhead to allow the most lightweight operation possible. Effective in a range of environmental conditions with high accuracy and very low impact on endpoint performance, StellarProtect™ provides full coverage in one deployment.

StellarEnforce™ optimizes operational efficiency for mass deployment and maintenance in large factories while reducing the chance of downtime and operational cost. StellarEnforce™ protects fixed-function and

legacy devices from malware infection or unauthorized changes, including industrial control systems (ICS), HMIs, SCADA, points of sale (POS), ATMs, and other embedded systems. StellarEnforce™ was specially designed to secure your legacy systems – it does this through its lockdown function, which limits execution of applications only to those that are necessary for daily operations. Through this technology StellarEnforce prevents cyber incidents without relying on pattern files.

Table 4: Example of Endpoint Solutions for SEMI E187

Key Solution		Description	SEMI E187 Requirement
1	Next Generation Malware Detection	Advanced threat scanning brings known attacks to heel while its next-gen machine learning engine blocks unknown threats, requiring no internet access.	Antivirus software enabled on equipment
2	Application Trust Listing	It does this through its lockdown function, which limits execution of applications only to those that are necessary for daily operations.	Equipment can disable unnecessary capabilities

- **Network Defense: The Edge Series**

EdgeIPS™ Pro is the leading Internal Segmentation IPS (ISIPS), designed to provide OT cybersecurity for large-scale highly-automated factories and empower smooth maintenance by IT security teams and OT security experts. Using EdgeIPS™ micro-segmentation technology as a building block, the Pro edition of EdgeIPS™ was designed specifically for environments requiring multiple network segments. The swappable, extensible module card allows for flexible deployment based on business intention, thus creating ideal network segmentation without altering existing network architecture. The high level of in-depth sensitivity to OT protocols allows administrators to design a scenario-based network trust list for machine-to-machine access control, while the virtual patch function offers asset owners the security shields they need to protect vulnerable OT assets and networks.

OT Defense Console™ (ODC) grants total OT visibility and the ability to deliver instant adjustment as necessary to protect your production line and keep the operation running.

Table 5: Example of Network Solutions for SEMI E187

Key Solution		Description	SEMI E187 Requirement
1	OT Protocol Filter Inspection	Edge solutions are able to inspect and create policy for application traffic	Support secure network protocol
2	Virtual Patching	Virtual patching shields vulnerable assets from network attacks leveraging known vulnerabilities	Have an alternative to security updates
3	Network Configuration Management	Streamline network access, establish security policy, and define your baseline with OT-native machine learning for network behavior	Provide a method to disable unnecessary network ports and protocols
4	Full Visibility of Assets, Operations, and Security Events	Full visibility of assets, operations, and security events	Support security monitoring
5	Network Segmentation	Network segmentation separate the ICS network's topology into different zones	Advanced feature

Conclusion

The SEMI E187 regulations provide essential guidance to asset owners seeking to secure their operation dependably and at low resource cost. With SEMI E187 it is possible for asset owners to mitigate OT cyber risks that were previously considered unmanageable. For equipment vendors, SEMI E187 also provides basic cybersecurity requirements that enhance system hardening and malware prevention.

TXOne's approach helps the semiconductor supply chain comply with these new standards with cyber defensive solutions based on the asset life cycle, including existing asset inspection, endpoint protection, and network defense. TXOne's specialists strongly recommend that organizations adopting the SEMI E187 standard adopt more automatic and comprehensive solutions to strengthen defenses and protect supply chain security.

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